



Moove Education Platform

Software Requirements Specification (SRS)

Version 3.1

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Prepared By:

Moove

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0.1 Document Management and Quality Assurance

0.1.1 Document Purpose and Scope

This Software Requirements Specification (SRS) serves as the authoritative source of system requirements for the Moove Education Platform—an Open edX-based learning management system deployed using Tutor. This document is a living artifact maintained in a Git repository, subject to version control, peer review, and continuous improvement.

0.1.2 Document Management

- **Version Control:** All documents are managed in a Git repository (`git@github.com:Moove-Digital-Solutions/moove-openedx-docs.git`)
- **Branching Strategy:** Main branch contains production-ready documentation; feature branches are used for improvements
- **Review Process:** All changes require peer review before merging
- **Release Process:** Documentation versions align with software releases

0.1.3 Tools and Methodology

- **Documentation Tool:** LaTeX with a custom preamble for consistent styling
- **Diagramming:** PlantUML for UML diagrams, with PNG exports for inclusion
- **Screenshots:** Captured from the production environment
- **Build Automation:** `build.sh` script compiles all documents with `latexmk`
- **Continuous Improvement:** Regular reviews against INSA regulations

0.1.4 Compliance Framework

This document and the system it describes are designed to comply with:

- Critical Mass Cyber Security Requirement Standard (CMCSRS) v2.0
- FDRE National Cybersecurity Policy (2024)
- Computer Crime Proclamation No. 958/2016
- Electronic Signature Proclamation No. 1072/2018
- Cyber Security Audit and Evaluation Guideline v1.0
- Cyber Security Risk Assessment Framework v1.0

0.1.5 Quality Assurance Measures

- **Automated Validation:** The `assess_security.sh` script validates security controls
- **Vulnerability Scanning:** Trivy performs daily container image scanning
- **Log Review:** Centralized logging via Tutor log analysis
- **Health Checks:** Container health checks configured for all services

0.2 Introduction

0.2.1 Purpose

This document defines the functional and non-functional requirements for the Moove Education Platform, an Open edX-based learning management system deployed using Tutor. It serves as the living source of truth for system capabilities, guiding continuous development, internal quality assurance, and security assessment activities. The requirements are aligned with Ethiopian regulatory frameworks, including the Critical Mass Cyber Security Requirement Standard (CMCSRS), the National Cybersecurity Policy, the Computer Crime Proclamation, and the Electronic Signature Proclamation.

The SRS has been developed following IEEE Std 830-1998 guidelines for software requirements specifications, ensuring completeness, consistency, and traceability.

0.2.2 Scope

The system provides a comprehensive online learning environment supporting:

- Course creation, authoring, and management (Educator features).
- Learner enrollment, progression, and engagement (Learner features).
- Content delivery (video, text, interactive components).
- Assessments (quizzes, assignments, peer reviews, proctored exams).
- Collaboration (discussions, teams, wikis).
- Analytics and reporting.
- Administrative controls.

0.2.3 System Overview and Actors

The platform is a client-server application based on Open edX, deployed in Docker containers orchestrated by Tutor. It uses a microservices architecture with separate containers for LMS, CMS, MFE, databases, and supporting services.

Actors

- **Educator (Instructor/Staff):** Creates and manages courses, creates content, assesses learners, moderates discussions, views analytics.
- **Learner (Student):** Enrolls in courses, accesses content, completes assessments, participates in discussions, tracks progress.
- **Administrator:** Manages users, courses, site configuration, and system health.
- **System:** Automated processes (notifications, email, background tasks).

0.2.4 System Constraints

- The system must comply with Ethiopian laws and regulations, including the Computer Crime Proclamation and the Electronic Signature Proclamation.
- All user data must be stored and processed within the country's jurisdiction.
- The system must support English and Amharic (planned) interfaces.
- The system shall be accessible via modern web browsers and mobile devices.

0.2.5 Assumptions and Dependencies

- The platform assumes reliable internet connectivity for both learners and educators.
- Third-party services (e.g., video hosting, proctoring) are available and secured.
- The underlying Tutor deployment and container infrastructure are maintained and updated regularly.

0.3 Functional Requirements – Educator Features

0.3.1 Course Creation & Planning Module

General Description Provides tools for educators to create, plan, and structure courses.

Requirement UFR-OE-001-01: Create a New Course

Overview Educators need a straightforward way to instantiate a new course shell without requiring technical intervention. This requirement ensures that course creation is guided, that the unique course identifier (composed of organisation, course number, and run) is validated, and that the system automatically sets up the necessary database records, audit logs, and default settings. This allows instructors to focus on content authoring rather than administrative setup.

1. **Title:** Create a new course.
2. **UML Traceability:** Use Case: “Create Course”; Activity: “Course Creation Flow”.

3. Actor: Educator (Instructor/Staff).

4. Pre-conditions:

- Educator is logged in with course creation permissions.
- The organisation code, course number, and run combination is unique.

5. Post-conditions:

- A new course skeleton is created in the CMS.
- The course appears in the educator's course list.
- An audit log entry is recorded.

6. Normal Flow:

- (a) Educator navigates to Studio (CMS).
- (b) Clicks "New Course".
- (c) Enters course title, organisation code, course number, and course run.
- (d) Sets start date, enrollment start/end dates, and course visibility.
- (e) Clicks "Create".
- (f) System validates uniqueness and creates the course ID.

7. Alternative Flows:

- Alt 1: Duplicate course ID → validation error.

8. Business Rules: Course ID format: org+course+run; must be unique.

9. Input/Output: Input: title, org, number, run, start date, enrollment dates. Output: Course object with ID.

10. Acceptance Criteria:

```
Given I am an educator
When I create a course with title "Introduction to Physics"
Then I receive a 201 Created
And the course appears in my course list
```

Requirement UFR-OE-001-02: Plan a New Course (Outline)

Overview Once a course shell exists, the educator must define its structure. This requirement covers the creation of a hierarchical outline using sections, subsections, and units. A clear outline helps learners navigate the material logically and enables the educator to organise content progressively. Reordering via drag-and-drop adds flexibility for iterative design.

1. Title: Plan course outline.

2. UML Traceability: Activity: "Course Outline Flow".

3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Course outline (sections, subsections, units) is defined.
6. **Normal Flow:**
 - (a) Educator opens the course outline.
 - (b) Clicks “Add Section” and enters a name.
 - (c) Within a section, clicks “Add Subsection” and enters a name.
 - (d) Within a subsection, clicks “Add Unit” to create content pages.
 - (e) Drags and drops to reorder.
7. **Alternative Flows:** None.
8. **Business Rules:** Sections, subsections, and units have a hierarchical order; re-ordering is allowed.
9. **Input/Output:** Input: section/subsection/unit metadata. Output: updated course structure.
10. **Acceptance Criteria:**

Given I am an educator with a course
When I add a section named "Week 1"
Then the section appears in the course outline
And I can add subsections under it

Requirement UFR-OE-001-03: Enroll Learners in a Course

Overview Educators may need to manually add learners to a course, especially for closed cohorts or corporate training. This requirement provides both individual and bulk enrollment options, along with email confirmation to keep learners informed. The system must validate email addresses and enrollment modes to ensure data integrity.

1. **Title:** Enroll learners manually.
2. **UML Traceability:** Use Case: “Manage Enrollments”.
3. **Actor:** Educator.
4. **Pre:** Course exists and enrollment is open.
5. **Post:** Learner is enrolled.
6. **Normal Flow:**
 - (a) Educator goes to “Instructor Dashboard” → “Enrollment”.
 - (b) Enters learner email(s) and selects enrollment track (audit/verified).
 - (c) Clicks “Enroll”.

(d) System sends confirmation email.

7. Alternative Flows: Bulk enrollment via CSV upload.

8. Business Rules: CSV format: `email,enrollment_mode`; only valid emails are processed.

9. Input/Output: Input: email(s), enrollment mode. Output: enrollment confirmation.

10. Acceptance Criteria:

Given I am an educator

When I enroll a learner with email `student@example.com` in verified mode

Then the learner is enrolled and receives a confirmation email

0.3.2 Instructional Design Concepts

Requirement UFR-OE-002-01: Apply Instructional Design Principles

Overview Beyond technical features, the platform should encourage sound pedagogical practices. This guideline reminds educators to structure their courses with clear learning objectives, align assessments to those objectives, and follow proven models such as backward design and Bloom's taxonomy. Adherence to these principles improves learner outcomes and course quality.

1. Title: Apply instructional design concepts.

2. UML Traceability: – (guideline).

3. Actor: Educator.

4. Pre: Course exists.

5. Post: Course is structured using sound pedagogical design.

6. Normal Flow:

(a) Educator defines learning objectives for each section.

(b) Creates assessments aligned with objectives.

(c) Uses Open edX best practices (backward design, Bloom's taxonomy).

7. Alternative Flows: None.

8. Business Rules: None.

9. Input/Output: None; this is a guideline.

10. Acceptance Criteria: The course structure follows backward design and includes assessments at appropriate levels.

0.3.3 Learner Engagement & Communication Module

Requirement UFR-OE-003-01: Course Discussions

Overview Discussion forums are a critical component of collaborative learning. Educators need to enable and configure forum topics, moderation rules, and visibility settings. This requirement ensures that the discussion feature can be activated per course and that the instructor can set up a structure that encourages meaningful interaction.

1. **Title:** Configure course discussions.
2. **UML Traceability:** Use Case: “Manage Discussions”.
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Discussion forums enabled.
6. **Normal Flow:**
 - (a) Educator goes to “Advanced Settings” and enables discussions.
 - (b) Configures topics and moderation policies.
7. **Alternative Flows:** None.
8. **Business Rules:** Discussions are enabled only if the platform supports them.
9. **Input/Output:** Input: enable flag, topics. Output: updated course settings.
10. **Acceptance Criteria:**

Given I am an educator
When I enable discussions and set topics
Then the Discussions tab appears in the course navigation

Requirement UFR-OE-003-02: Bulk Emails

Overview Effective communication with large learner cohorts often requires sending announcements, reminders, or personalised messages. This requirement enables educators to compose and send bulk emails to all or segmented groups of enrolled learners. The system handles queuing and delivery asynchronously to avoid performance bottlenecks.

1. **Title:** Send bulk emails to learners.
2. **UML Traceability:** Sequence: “Bulk Email Flow”.
3. **Actor:** Educator.
4. **Pre:** Course has enrolled learners.
5. **Post:** Email sent asynchronously.
6. **Normal Flow:**

- (a) Educator composes email, selects recipients (all learners, specific segments).
- (b) Clicks “Send”.
- (c) System queues via Celery.

7. Alternative Flows: None.

8. Business Rules: Emails are sent in batches; rate limits apply.

9. Input/Output: Input: subject, body, recipient list. Output: task ID.

10. Acceptance Criteria:

```
Given I am an educator
When I send a bulk email to all learners
Then the email is queued and delivered within 5 minutes
```

Requirement UFR-OE-003-03: Course Updates

Overview Educators frequently need to post time-sensitive news, clarifications, or motivational messages on the course home page. This requirement allows them to create and publish short updates that are immediately visible to all enrolled learners, keeping the community informed and engaged.

1. Title: Publish course updates.

2. UML Traceability: – (simple notification).

3. Actor: Educator.

4. Pre: Course exists.

5. Post: Update appears on course home page.

6. Normal Flow: Educator creates an update post (title, content) and publishes.

7. Alternative Flows: None.

8. Business Rules: Updates are visible to all enrolled learners.

9. Input/Output: Input: title, content. Output: published update.

10. Acceptance Criteria:

```
Given I am an educator
When I publish a course update
Then it appears on the course home page immediately
```

0.3.4 Content Creation & Management Module

Requirement UFR-OE-004-01: Create and Manage Components

Overview At the core of course authoring is the ability to add diverse learning materials – videos, HTML texts, problems, discussions, and more – to any unit. This requirement provides a unified interface for selecting component types, configuring their content, and saving them into the course structure. Each component type leverages the XBlock framework for extensibility.

1. **Title:** Add and manage course components.
2. **UML Traceability:** Activity: “Component Management”.
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** Component added to unit.
6. **Normal Flow:**
 - (a) Educator selects a unit.
 - (b) Clicks “Add New Component” and chooses type (HTML, Video, Problem, Discussion, etc.).
 - (c) Configures component content.
 - (d) Saves.
7. **Alternative Flows:** None.
8. **Business Rules:** Component types are defined by the XBlock system.
9. **Input/Output:** Input: component type, configuration. Output: component stored.
10. **Acceptance Criteria:**

Given I am an educator
When I add a Video component to a unit
Then the video appears in the unit content

Requirement UFR-OE-004-02: Manage Text Components

Overview Text components are the most basic but also the most versatile building block. Educators need a rich editing environment that supports formatting, images, LaTeX for mathematics, and code snippets. The system must also sanitize HTML to prevent cross-site scripting, ensuring safe rendering for all learners.

1. **Title:** Create and edit HTML/text components.
2. **Actor:** Educator.

3. **Pre:** Unit exists.
4. **Post:** Text component saved.
5. **Normal Flow:** Uses rich text editor or raw HTML; supports LaTeX, code snippets.
6. **Alternative Flows:** None.
7. **Business Rules:** HTML must be sanitized to prevent XSS.
8. **Input/Output:** Input: HTML content. Output: stored component.
9. **Acceptance Criteria:**

```
Given I am an educator
When I edit a text component with LaTeX
Then the rendered content displays the math correctly
```

Requirement UFR-OE-004-03: Manage Video Components

Overview Video is a primary medium for online courses. Educators need to upload video files or embed from external sources (YouTube, Vimeo) and optionally attach transcripts. The video player must provide standard controls (play, speed, captions) and support adaptive bitrate streaming for varied network conditions.

1. **Title:** Upload and manage video content.
2. **UML Traceability:** Use Case: “Manage Video”.
3. **Actor:** Educator.
4. **Pre:** Video file available.
5. **Post:** Video player embedded.
6. **Normal Flow:** Uploads video file or enters URL; system processes and displays player with transcripts, speed control.
7. **Alternative Flows:** URL from YouTube/Vimeo.
8. **Business Rules:** Video must be in supported formats; transcripts are optional.
9. **Input/Output:** Input: video file or URL, transcript (optional). Output: video player.
10. **Acceptance Criteria:**

```
Given I am an educator
When I upload a video file
Then the video player appears and plays the video
```

Requirement UFR-OE-004-04: Control Content Visibility & Access

Overview Not all content should be available immediately; educators often schedule release dates, set prerequisites, or restrict access to specific groups. This requirement provides fine-grained visibility controls so that content can be rolled out in a controlled manner, supporting just-in-time learning and course pacing.

1. **Title:** Control content visibility and access.
2. **UML Traceability:** – (settings).
3. **Actor:** Educator.
4. **Pre:** Content exists.
5. **Post:** Visibility rules applied.
6. **Normal Flow:** Sets release dates, prerequisites, and visibility rules.
7. **Alternative Flows:** None.
8. **Business Rules:** Release dates must be in the future; prerequisites must be valid content IDs.
9. **Input/Output:** Input: release date, prerequisites. Output: updated content visibility.
10. **Acceptance Criteria:**

```
Given I am an educator
When I set a release date in the future
Then the content is hidden until that date
```

0.3.5 Components & Activities Module

Requirement UFR-OE-005-01: Dropdown Problem

Overview Single-select dropdown problems are useful for concise assessments where space is limited. Educators can define a question, a list of options, and mark the correct answer. The system automatically grades the learner's selection and provides immediate feedback.

1. **Title:** Create dropdown problem.
2. **UML Traceability:** – (component type).
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** Problem added to unit.
6. **Normal Flow:**

- (a) Adds a Problem component.
- (b) Selects “Dropdown” type.
- (c) Enters question and options.
- (d) Marks correct answer.
- (e) Saves.

7. Alternative Flows: None.

8. Business Rules: One correct answer.

9. Input/Output: Input: question, options, correct answer. Output: problem stored.

10. Acceptance Criteria:

```
Given I am an educator
When I create a dropdown problem
Then the problem is displayed and grades the learner's selection
```

Requirement UFR-OE-005-02: Multi-Select Problem

Overview Multi-select problems allow learners to choose multiple correct answers from a list. This format is ideal for questions with more than one correct option. Educators can define partial credit rules to reward partially correct selections, making assessments more nuanced.

1. Title: Create multi-select problem.

2. Actor: Educator.

3. Pre: Unit exists.

4. Post: Problem added.

5. Normal Flow: Adds Problem, selects “Multi-Select”, enters options and marks multiple correct answers.

6. Alternative Flows: None.

7. Business Rules: Multiple correct answers; partial credit possible.

8. Input/Output: Input: question, options, correct options. Output: problem stored.

9. Acceptance Criteria:

```
Given I am an educator
When I create a multi-select problem
Then the problem allows selection of multiple options and grades accordingly
```

Requirement UFR-OE-005-03: Open Response Assessments (ORA)

Overview Open Response Assessments go beyond simple multiple-choice questions by enabling learners to submit free-text responses and participate in peer review. Educators can define a prompt, a detailed rubric, and configure the assessment workflow (self, peer, staff). This promotes higher-order thinking and collaborative learning.

1. **Title:** Create open response assessment.
2. **UML Traceability:** Activity: “ORA Flow”.
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** ORA created.
6. **Normal Flow:**
 - (a) Adds Advanced component → “Open Response Assessment”.
 - (b) Defines prompt and rubric.
 - (c) Configures assessment steps (self, peer, staff).
 - (d) Sets deadlines.
 - (e) Saves.
7. **Alternative Flows:** None.
8. **Business Rules:** Rubric criteria defined; deadlines must be in order.
9. **Input/Output:** Input: prompt, rubric, steps, deadlines. Output: ORA stored.
10. **Acceptance Criteria:**

Given I am an educator
When I create an ORA with peer assessment
Then learners can submit responses and review peers

Requirement UFR-OE-005-04: Single Select Problem

Overview Single-select (radio button) problems are straightforward for yes/no or one-correct-answer questions. This requirement ensures that educators can quickly add this assessment type, with the system handling grading and feedback consistently.

1. **Title:** Create single select (radio button) problem.
2. **Actor:** Educator.
3. **Pre:** Unit exists.
4. **Post:** Problem added.

5. **Normal Flow:** Adds Problem, selects “Single Select”, enters options, marks correct answer.
6. **Alternative Flows:** None.
7. **Business Rules:** One correct answer.
8. **Input/Output:** Input: question, options, correct answer. Output: problem stored.
9. **Acceptance Criteria:**

```
Given I am an educator
When I create a single select problem
Then the problem displays radio buttons and grades accordingly
```

Requirement UFR-OE-005-05: Text Input Problem

Overview Text input problems require learners to type a short answer. Educators define the expected answer(s) and the system checks the input (case-insensitive). This format is ideal for fill-in-the-blank or short-answer questions, providing immediate correctness feedback.

1. **Title:** Create text input problem.
2. **Actor:** Educator.
3. **Pre:** Unit exists.
4. **Post:** Problem added.
5. **Normal Flow:** Adds Problem, selects “Text Input”, enters question and expected answer(s).
6. **Alternative Flows:** None.
7. **Business Rules:** Supports multiple accepted answers (case-insensitive).
8. **Input/Output:** Input: question, expected answers. Output: problem stored.
9. **Acceptance Criteria:**

```
Given I am an educator
When I create a text input problem
Then the learner can type an answer and receive immediate feedback
```

0.3.6 Data & Analytics Module

Requirement UFR-OE-006-01: View Course Enrollments

Overview Educators need insights into enrollment trends to gauge course popularity and plan capacity. This requirement provides access to real-time enrollment statistics and downloadable CSV reports, enabling data-driven decisions about marketing, outreach, and resource allocation.

1. **Title:** View course enrollment data.
2. **UML Traceability:** Use Case: “View Analytics”.
3. **Actor:** Educator.
4. **Pre:** Educator has instructor access.
5. **Post:** Enrollment report displayed.
6. **Normal Flow:** Accesses Instructor Dashboard → Data Downloads → Enrollment Data; views totals, trends, demographics.
7. **Alternative Flows:** None.
8. **Business Rules:** Data is real-time.
9. **Input/Output:** Output: CSV with enrollment details.
10. **Acceptance Criteria:**

```
Given I am an educator
When I download enrollment data
Then I receive a CSV with all active enrollments
```

Requirement UFR-OE-006-02: View Learner Data

Overview To support individual learners effectively, educators must be able to look up a specific learner’s progress, engagement metrics, and activity history. This requirement provides a search interface that returns a comprehensive profile, enabling personalised intervention and support.

1. **Title:** View learner data and activity.
2. **Actor:** Educator.
3. **Pre:** Educator has instructor access.
4. **Post:** Learner information displayed.
5. **Normal Flow:** Instructor Dashboard → Student Admin; searches learner; views progress, engagement, activity history.
6. **Alternative Flows:** None.

7. **Business Rules:** Data is anonymized if required.
8. **Input/Output:** Input: learner identifier. Output: learner profile with activity.
9. **Acceptance Criteria:**

```
Given I am an educator
When I search for a learner
Then I see their progress and activity logs
```

Requirement UFR-OE-006-03: View Learner Grades

Overview The gradebook is the central hub for monitoring learner performance. Educators can view grades per assignment, adjust them manually if needed, and export the entire gradebook to CSV for offline analysis. All grade changes are audited to maintain integrity.

1. **Title:** View and manage learner grades.
2. **UML Traceability:** Sequence: “Grade View”.
3. **Actor:** Educator.
4. **Pre:** Educator has instructor access.
5. **Post:** Gradebook displayed.
6. **Normal Flow:** Accesses Gradebook; views grades per assignment; adjusts manually if needed; exports CSV.
7. **Alternative Flows:** None.
8. **Business Rules:** Grade changes are logged.
9. **Input/Output:** Output: gradebook; input: grade adjustments.
10. **Acceptance Criteria:**

```
Given I am an educator
When I adjust a grade in the gradebook
Then the grade is updated and an audit log is created
```

Requirement UFR-OE-006-04: View Answer Data

Overview Understanding which problems are difficult for learners helps educators improve course content. This requirement provides aggregated answer distribution statistics, showing the percentage of correct answers per problem. This data informs decisions about revising questions or adding remedial material.

1. **Title:** View answer distribution data.
2. **Actor:** Educator.

3. **Pre:** Educator has instructor access.
4. **Post:** Distribution displayed.
5. **Normal Flow:** Data Downloads → Answer Distribution; sees % correct per problem.
6. **Alternative Flows:** None.
7. **Business Rules:** Data is aggregated.
8. **Input/Output:** Output: distribution report.
9. **Acceptance Criteria:**

```
Given I am an educator
When I view answer distribution
Then I see the percentage of correct answers for each problem
```

Requirement UFR-OE-006-05: View Certificate Data

Overview Educators need to track certificate issuance to confirm completion and verify learner achievements. This requirement provides a report of all issued certificates, including learner names, URLs, and generation dates, supporting both internal review and external verification requests.

1. **Title:** View certificate issuance data.
2. **Actor:** Educator.
3. **Pre:** Educator has instructor access.
4. **Post:** Certificate data displayed.
5. **Normal Flow:** Data Downloads → Certificate Data; views list of issued certificates and download stats.
6. **Alternative Flows:** None.
7. **Business Rules:** Data includes learner names, certificate URLs, and generation dates.
8. **Input/Output:** Output: certificate report.
9. **Acceptance Criteria:**

```
Given I am an educator
When I download certificate data
Then I get a list of all issued certificates
```

0.3.7 Accessibility Module

Requirement UFR-OE-007-01: Open edX Accessibility Guidelines

Overview The platform must be usable by all learners, including those with disabilities. This requirement mandates that educators follow Open edX's accessibility guidelines, which align with WCAG 2.1 AA standards. Compliance is verified during course audits, ensuring equal access to learning.

1. **Title:** Ensure accessibility compliance.
2. **Actor:** Educator.
3. **Pre:** Course content being developed.
4. **Post:** Content meets WCAG 2.1 AA.
5. **Normal Flow:** Educator follows guidelines; uses alt text, transcripts, proper structure.
6. **Alternative Flows:** None.
7. **Business Rules:** Compliance is checked during audit.
8. **Input/Output:** None.
9. **Acceptance Criteria:** All content passes accessibility checks.

Requirement UFR-OE-007-02: Mobile Experience Design

Overview Many learners access courses via mobile devices. This requirement ensures that all content is responsive and functions correctly on iOS and Android browsers. Educators are encouraged to test their materials on mobile and adjust interactions (e.g., touch targets, font sizes) to provide a seamless experience.

1. **Title:** Design for mobile experience.
2. **Actor:** Educator.
3. **Pre:** Course content created.
4. **Post:** Content is responsive and functional on mobile.
5. **Normal Flow:** Educator tests content on mobile devices; adjusts interactions.
6. **Alternative Flows:** None.
7. **Business Rules:** Must work on iOS and Android.
8. **Input/Output:** None.
9. **Acceptance Criteria:** Course renders correctly on mobile browsers.

Requirement UFR-OE-007-03: Accessibility Best Practices Checklist

Overview To simplify adherence to accessibility standards, this requirement provides a checklist that educators can review and confirm for each course. The checklist covers alt text, heading structure, colour contrast, keyboard navigation, and descriptive links – all items that contribute to an inclusive learning environment.

1. **Title:** Follow accessibility best practices.
2. **Actor:** Educator.
3. **Pre:** Content created.
4. **Post:** Checklist items satisfied.
5. **Normal Flow:** Educator reviews checklist (alt text, headings, contrast, keyboard navigation, descriptive links).
6. **Alternative Flows:** None.
7. **Business Rules:** All items must be satisfied.
8. **Input/Output:** None.
9. **Acceptance Criteria:** Checklist is completed and verified.

Requirement UFR-OE-007-04: Manage Video Transcripts

Overview Closed captions are essential for hearing-impaired learners and beneficial for all in noisy environments. This requirement enables educators to upload transcript files (SRT/VTT) for any video component, which then appear as captions in the video player, improving accessibility and comprehension.

1. **Title:** Upload and manage video transcripts.
2. **Actor:** Educator.
3. **Pre:** Video component exists.
4. **Post:** Transcript attached.
5. **Normal Flow:** Uploads .srt/.vtt files for video components; transcripts appear as captions.
6. **Alternative Flows:** None.
7. **Business Rules:** Transcript must be in supported format.
8. **Input/Output:** Input: transcript file. Output: captions enabled.
9. **Acceptance Criteria:**

Given I am an educator
When I upload a transcript for a video
Then the video player shows closed captions

0.3.8 Import/Export / OLX Module

Requirement UFR-OE-008-01: Open Learning XML (OLX) Support

Overview OLX is the standard exchange format for Open edX courses. This requirement ensures that educators can export a course as an OLX archive (for backup, versioning, or transfer) and import an archive to recreate a course. This interoperability is vital for collaboration and disaster recovery.

1. **Title:** Support for Open Learning XML.
2. **Actor:** Educator.
3. **Pre:** Course exists.
4. **Post:** Course can be exported/imported.
5. **Normal Flow:** Educator uses export/import features; system generates or consumes OLX archives.
6. **Alternative Flows:** None.
7. **Business Rules:** OLX must be valid.
8. **Input/Output:** Input: OLX archive; output: exported archive.
9. **Acceptance Criteria:**

```
Given I am an educator
When I export a course
Then I receive a .tar.gz file containing OLX files
When I import the same file
Then the course structure is recreated exactly
```

Requirement UFR-OE-008-02: Getting Started with OLX

Overview Educators new to OLX need accessible documentation and tools to understand the format. This requirement ensures that the platform provides links to official Open edX OLX documentation, command-line utilities, and examples, lowering the barrier to using advanced import/export features.

1. **Title:** Access OLX documentation and tools.
2. **Actor:** Educator.
3. **Pre:** None.
4. **Post:** Educator can learn OLX.
5. **Normal Flow:** System provides links to documentation and command-line tools.
6. **Alternative Flows:** None.
7. **Business Rules:** Documentation must be current.

8. **Input/Output:** None.
9. **Acceptance Criteria:** Documentation is accessible from the platform.

0.4 Functional Requirements – Learner Features

0.4.1 Dashboard & Profile Module

Requirement UFR-OE-101-01: Explore Dashboard and Profile

Overview After logging in, learners are greeted by a personalised dashboard that aggregates their enrolled courses, progress summaries, announcements, and recent activity. This central hub provides an at-a-glance view of their learning journey, making it easy to resume courses or discover new opportunities.

1. **Title:** Access and explore learner dashboard.
2. **UML Traceability:** Use Case: “View Dashboard”.
3. **Actor:** Learner.
4. **Pre:** Learner logged in.
5. **Post:** Dashboard displays enrolled courses, progress, announcements.
6. **Normal Flow:**
 - (a) Learner logs in.
 - (b) Dashboard shows course cards with progress bars, recent activity, and site announcements.
7. **Alternative Flows:** None.
8. **Business Rules:** Dashboard is personalized.
9. **Input/Output:** Output: dashboard view.
10. **Acceptance Criteria:**

Given I am a learner
When I log in
Then I see my enrolled courses and progress

Requirement UFR-OE-101-02: Access Courses from Dashboard

Overview The dashboard presents each enrolled course as a card. Clicking on a card should take the learner directly into the course home page, where they can resume learning. This requirement ensures a seamless entry point into course content.

1. **Title:** Access enrolled courses from dashboard.
2. **Actor:** Learner.

3. **Pre:** Learner logged in.
4. **Post:** Course opens.
5. **Normal Flow:** Clicks on course card; enters the course.
6. **Alternative Flows:** None.
7. **Business Rules:** None.
8. **Input/Output:** Input: click on card. Output: course home page.
9. **Acceptance Criteria:**

```
Given I am a learner
When I click on a course card
Then I am taken to the course home page
```

Requirement UFR-OE-101-03: Share Courses on Social Media

Overview Learners often want to share interesting courses with their network. This requirement provides a share button that, when clicked, opens a social media sharing dialog (Facebook, Twitter, LinkedIn, etc.) with a pre-populated link to the course, facilitating organic promotion.

1. **Title:** Share course links on social media.
2. **Actor:** Learner.
3. **Pre:** Course page open.
4. **Post:** Link shared.
5. **Normal Flow:** Clicks “Share” button; selects platform; link posted.
6. **Alternative Flows:** None.
7. **Business Rules:** Must use standard sharing APIs.
8. **Input/Output:** Input: share action. Output: social media post.
9. **Acceptance Criteria:**

```
Given I am a learner
When I share a course on Facebook
Then the course link appears on my timeline
```

Requirement UFR-OE-101-04: Update Profile Image

Overview Learners can personalise their profile by uploading a profile picture. The system resizes and stores the image, ensuring it meets size and format constraints. A recognisable profile image helps build community and trust among peers.

1. **Title:** Upload or update profile image.
2. **Actor:** Learner.
3. **Pre:** Learner logged in.
4. **Post:** Image updated.
5. **Normal Flow:** Clicks on profile, selects image file; system resizes and stores.
6. **Alternative Flows:** None.
7. **Business Rules:** Image must be under 5MB; supported formats: PNG, JPG.
8. **Input/Output:** Input: image file. Output: updated profile.
9. **Acceptance Criteria:**

```
Given I am a learner
When I upload a new profile image
Then my profile displays the new image
```

Requirement UFR-OE-101-05: Update Profile Details

Overview Learners can maintain their personal information (name, email, location, bio, language) through a profile editing interface. The system enforces uniqueness of email addresses and validates input, ensuring data quality across the platform.

1. **Title:** Update profile information.
2. **Actor:** Learner.
3. **Pre:** Learner logged in.
4. **Post:** Profile updated.
5. **Normal Flow:** Edits fields (name, email, location, bio, language) and saves.
6. **Alternative Flows:** None.
7. **Business Rules:** Email must be unique.
8. **Input/Output:** Input: updated fields. Output: saved profile.
9. **Acceptance Criteria:**

```
Given I am a learner
When I change my bio and save
Then my profile shows the new bio
```

Requirement UFR-OE-101-06: Add Social Media Links

Overview To facilitate networking and professional connections, learners can add links to their social media profiles (Twitter, LinkedIn, GitHub, etc.). These appear as icons on their public profile, enabling others to connect outside the platform.

1. **Title:** Add personal social media links.
2. **Actor:** Learner.
3. **Pre:** Profile edit page open.
4. **Post:** Links added.
5. **Normal Flow:** Adds URLs for Twitter, LinkedIn, GitHub, etc.
6. **Alternative Flows:** None.
7. **Business Rules:** URLs must be valid.
8. **Input/Output:** Input: URLs. Output: icons on profile.
9. **Acceptance Criteria:**

```
Given I am a learner
When I add my Twitter URL
Then the Twitter icon appears on my profile
```

Requirement UFR-OE-101-07: Toggle Profile Visibility

Overview Privacy is important. This requirement allows learners to control who can see their profile information – either making it public (visible to all logged-in users) or private (visible only to themselves). The default is private, giving learners control over their data.

1. **Title:** Control profile visibility.
2. **Actor:** Learner.
3. **Pre:** Profile edit page open.
4. **Post:** Visibility updated.
5. **Normal Flow:** Toggles visibility of profile sections (public/private).
6. **Alternative Flows:** None.
7. **Business Rules:** Default is private.
8. **Input/Output:** Input: toggle setting. Output: updated visibility.
9. **Acceptance Criteria:**

```
Given I am a learner
When I set my profile to public
Then other users can see my profile information
```

Requirement UFR-OE-101-08: View Another Learner's Profile

Overview In collaborative courses, learners may want to see more about their peers. This requirement enables viewing another learner's public profile (if they have set it to public) by clicking on their username in discussions or the course roster. Only public information is shown.

1. **Title:** View other learners' profiles.
2. **Actor:** Learner.
3. **Pre:** Another learner's username is visible.
4. **Post:** Profile displayed.
5. **Normal Flow:** Clicks on username in discussions or roster; profile opens (public info only).
6. **Alternative Flows:** None.
7. **Business Rules:** Only public information is shown.
8. **Input/Output:** Input: username. Output: profile view.
9. **Acceptance Criteria:**

```
Given I am a learner
When I click on another learner's username
Then I see their public profile information
```

Requirement UFR-OE-101-09: Update Course-Specific Settings

Overview Learners can customise their experience per course – for example, disabling email notifications or choosing a preferred language. This requirement provides a settings panel within each course, allowing learners to tailor their notification preferences and other options without affecting other courses.

1. **Title:** Update course-specific settings.
2. **Actor:** Learner.
3. **Pre:** Enrolled in a course.
4. **Post:** Settings updated.
5. **Normal Flow:** In course, clicks avatar → Course Settings; adjusts email notifications, language.
6. **Alternative Flows:** None.
7. **Business Rules:** Settings are per course.
8. **Input/Output:** Input: settings changes. Output: saved.

9. Acceptance Criteria:

Given I am a learner enrolled in a course
When I disable email notifications
Then I no longer receive course emails

0.4.2 Course Navigation Module

Requirement UFR-OE-102-01: View Course Sections from Navigation Sidebar

Overview A clear navigation sidebar helps learners move through the course material intuitively. This requirement ensures that the sidebar displays all sections and subsections, and that clicking on any item takes the learner directly to that content. Only published and released content is visible, preventing confusion.

1. **Title:** Navigate course using sidebar.
2. **Actor:** Learner.
3. **Pre:** Enrolled in a course.
4. **Post:** Navigation displays sections.
5. **Normal Flow:** Sidebar displays all sections/subsections; click to navigate.
6. **Alternative Flows:** None.
7. **Business Rules:** Only published content is visible.
8. **Input/Output:** Input: click on item. Output: content displayed.
9. **Acceptance Criteria:**

Given I am a learner in a course
When I click on a section in the sidebar
Then I am taken to that section's content

0.4.3 Enrollment Module

Requirement UFR-OE-103-01: Enroll in a Course

Overview Enrolling in a course is the first step for any learner. This requirement provides a clear enrollment flow from the course catalog, allowing learners to choose between audit (free) and verified (paid) tracks. Upon confirmation, the course is added to the learner's dashboard.

1. **Title:** Enroll in a course.
2. **UML Traceability:** Sequence: "Enrollment Flow".
3. **Actor:** Learner.

4. **Pre:** Course open for enrollment.
5. **Post:** Learner enrolled.
6. **Normal Flow:**
 - (a) Browses course catalog.
 - (b) Clicks “Enroll Now”.
 - (c) Selects enrollment track (Audit/Verified).
 - (d) Confirms enrollment.
7. **Alternative Flows:** None.
8. **Business Rules:** Enrollment may require payment for verified track.
9. **Input/Output:** Input: course selection, track. Output: enrollment confirmation.
10. **Acceptance Criteria:**

Given I am a learner

When I enroll in a course

Then the course appears in my dashboard

Requirement UFR-OE-103-02: Open and Start a Course

Overview Once enrolled, learners need a clear way to begin or resume their learning. This requirement provides “Start Course” for first-time access and “Resume Course” to return to the last viewed unit, minimising friction and helping learners stay on track.

1. **Title:** Open and start a course.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Course opens to first content.
5. **Normal Flow:** Clicks “Start Course” or “Resume Course” from dashboard; goes to last visited content.
6. **Alternative Flows:** None.
7. **Business Rules:** Resume picks last visited unit.
8. **Input/Output:** Input: click start/resume. Output: course content.
9. **Acceptance Criteria:**

Given I am a learner enrolled in a course

When I click "Resume Course"

Then I am taken to the last content I viewed

0.4.4 Course Prerequisites Module

Requirement UFR-OE-104-01: Overview of Prerequisites

Overview Prerequisites help ensure learners have the necessary background before starting a course. This requirement displays a clear list of required courses on the course info page, along with the learner's completion status for each, so they can plan their learning path accordingly.

1. **Title:** View course prerequisites.
2. **Actor:** Learner.
3. **Pre:** Course info page.
4. **Post:** Prerequisites listed.
5. **Normal Flow:** Prerequisites displayed on course info page; learner can check completion status.
6. **Alternative Flows:** None.
7. **Business Rules:** Prerequisites are defined by educator.
8. **Input/Output:** Output: prerequisite list.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view a course with prerequisites
Then I see the required courses and my completion status
```

Requirement UFR-OE-104-02: Complete Prerequisite Course

Overview To automatically unlock a course, the learner must complete its prerequisites with a passing grade. This requirement ensures that completion is tracked and that the advanced course becomes available in the dashboard once all prerequisites are met.

1. **Title:** Complete prerequisite course.
2. **Actor:** Learner.
3. **Pre:** Prerequisite course defined.
4. **Post:** Prerequisite marked complete.
5. **Normal Flow:** Must complete prerequisite course before enrolling in advanced course.
6. **Alternative Flows:** None.
7. **Business Rules:** Completion is based on passing grade.
8. **Input/Output:** None.

9. Acceptance Criteria:

Given I am a learner
When I complete a prerequisite course with a passing grade
Then the advanced course becomes available

Requirement UFR-OE-104-03: Pass Entrance Exam

Overview Some courses may require an entrance exam to assess readiness. This requirement allows educators to set a minimum score, and learners can attempt the exam (with retake cooldown if they fail). Passing automatically grants access to the course content.

1. **Title:** Pass entrance exam.
2. **Actor:** Learner.
3. **Pre:** Course requires entrance exam.
4. **Post:** Exam passed.
5. **Normal Flow:** Takes and passes an entrance exam to gain access.
6. **Alternative Flows:** If fail, retake allowed after cooldown.
7. **Business Rules:** Minimum score set by educator.
8. **Input/Output:** Input: exam answers. Output: pass/fail.
9. **Acceptance Criteria:**

Given I am a learner
When I pass the entrance exam
Then I can access the course content

0.4.5 Course Content Access Module

Requirement UFR-OE-105-01: Troubleshoot Accessing Course Content

Overview When learners encounter access restrictions (e.g., locked content due to prerequisites or release dates), they need clear explanations and guidance. This requirement ensures that error messages are user-friendly, explain the reason, and provide links to support or course staff for assistance.

1. **Title:** Troubleshoot content access issues.
2. **Actor:** Learner.
3. **Pre:** Content not accessible.
4. **Post:** Issue resolved or guidance given.

5. **Normal Flow:** System provides error messages and support links; contact course staff.
6. **Alternative Flows:** None.
7. **Business Rules:** Error messages are user-friendly.
8. **Input/Output:** Output: error message.
9. **Acceptance Criteria:**

Given I am a learner
When I try to access locked content
Then I see a message explaining why and how to gain access

Requirement UFR-OE-105-02: Access an Archived Course

Overview After a course ends, it may be archived. Learners should still be able to view the content in read-only mode, but submissions and interactions are disabled. This requirement supports ongoing reference and revision without allowing new activity.

1. **Title:** Access archived course.
2. **Actor:** Learner.
3. **Pre:** Course archived.
4. **Post:** Read-only access.
5. **Normal Flow:** Views read-only archived course content.
6. **Alternative Flows:** None.
7. **Business Rules:** No submissions allowed.
8. **Input/Output:** Input: click on archived course. Output: read-only view.
9. **Acceptance Criteria:**

Given I am a learner
When I access an archived course
Then I can view content but not submit assignments

Requirement UFR-OE-105-03: View Final Course Grade

Overview At the end of a course, learners need to see their final grade and whether they passed. This requirement provides a dedicated “Grades” page that summarises all assignment scores and the overall final grade, giving learners a clear picture of their performance.

1. **Title:** View final course grade.

2. **Actor:** Learner.
3. **Pre:** Course completed.
4. **Post:** Grade displayed.
5. **Normal Flow:** Goes to course dashboard; clicks Grades; final grade displayed.
6. **Alternative Flows:** None.
7. **Business Rules:** Grade calculated from all assessments.
8. **Input/Output:** Output: final grade.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view my course grades
Then I see my final grade and passing status
```

Requirement UFR-OE-105-04: View Certificate Status

Overview Learners who pass a course can earn a certificate. This requirement shows the certificate status (eligible, available, or not eligible) in the profile, and provides a download link when available. This motivates learners and gives them a tangible credential.

1. **Title:** View certificate status.
2. **Actor:** Learner.
3. **Pre:** Course completed.
4. **Post:** Status displayed.
5. **Normal Flow:** In profile → Certificates, sees status (Eligible, Available, Not Eligible).
6. **Alternative Flows:** None.
7. **Business Rules:** Certificate available if passing grade.
8. **Input/Output:** Output: certificate status.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view my certificate status
Then I see if I am eligible and can download it
```

0.4.6 Course Progress Module

Requirement UFR-OE-106-01: Find Course Start Date

Overview Knowing the course start date helps learners plan their schedule. This requirement ensures that the start date is prominently displayed on the course info page and on the dashboard, so learners are aware of when the course officially begins.

1. **Title:** View course start date.
2. **Actor:** Learner.
3. **Pre:** Course info page.
4. **Post:** Start date displayed.
5. **Normal Flow:** Start date displayed on course info and dashboard.
6. **Alternative Flows:** None.
7. **Business Rules:** Start date set by educator.
8. **Input/Output:** Output: start date.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view a course
Then I see the start date
```

Requirement UFR-OE-106-02: Start or Resume a Course

Overview This requirement is similar to UFR-OE-103-02 but appears in the context of progress. It ensures that the “Start”/“Resume” button is always accessible from the course home page and the dashboard, allowing learners to jump directly to their last visited content.

1. **Title:** Start or resume a course.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Course content opened.
5. **Normal Flow:** Clicks “Start” or “Resume” from dashboard; goes to last visited content.
6. **Alternative Flows:** None.
7. **Business Rules:** Resume picks last unit.
8. **Input/Output:** Input: click. Output: content.

9. Acceptance Criteria:

```
Given I am a learner
When I click "Resume"
Then I am taken to the last unit I viewed
```

Requirement UFR-OE-106-03: Progress Indicators

Overview Visual progress indicators (e.g., progress bars) help learners understand how much of a section or subsection they have completed. This requirement provides these indicators at multiple levels – overall course, section, and subsection – to motivate and guide learners through the material.

1. **Title:** View progress indicators.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Progress shown.
5. **Normal Flow:** Visual indicators show completion percentage for sections, subsections, and overall.
6. **Alternative Flows:** None.
7. **Business Rules:** Progress is calculated automatically.
8. **Input/Output:** Output: progress bars.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view a course section
Then I see a progress bar indicating completion
```

Requirement UFR-OE-106-04: The Progress Page

Overview For a detailed view of progress, learners can access a dedicated Progress page that shows scores for each assignment and overall completion status. This page is particularly useful for tracking performance and identifying areas that need more attention.

1. **Title:** Access the Progress page.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Detailed progress displayed.
5. **Normal Flow:** Navigates to Progress page to view detailed completion status.

6. **Alternative Flows:** None.
7. **Business Rules:** Shows per-assignment scores.
8. **Input/Output:** Output: progress detail.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view the Progress page
Then I see scores for each assignment and overall completion
```

0.4.7 Certificates & Badges Module

Requirement UFR-OE-107-01: Accessing a Certificate

Overview Earned certificates are a key motivator. This requirement allows learners to download their certificate as a signed PDF from their profile. The certificate includes the learner's name, course title, and a unique verification code for authenticity.

1. **Title:** Access earned certificate.
2. **Actor:** Learner.
3. **Pre:** Certificate available.
4. **Post:** Certificate downloaded.
5. **Normal Flow:** Goes to profile → Certificates → download.
6. **Alternative Flows:** None.
7. **Business Rules:** Certificate is PDF with digital signature.
8. **Input/Output:** Input: click download. Output: PDF file.
9. **Acceptance Criteria:**

```
Given I am a learner with an available certificate
When I click download
Then a PDF certificate is generated and downloaded
```

Requirement UFR-OE-107-02: Viewing Earned Badges in Profile

Overview Badges recognise specific achievements (e.g., completing a module, scoring high on an assignment). This requirement displays all earned badges in the learner's profile, providing a visual record of accomplishments that can be shared with others.

1. **Title:** View earned badges in profile.
2. **Actor:** Learner.

3. **Pre:** Badges earned.
4. **Post:** Badges displayed.
5. **Normal Flow:** Badges appear in profile; click to view details.
6. **Alternative Flows:** None.
7. **Business Rules:** Badges are awarded based on achievements.
8. **Input/Output:** Output: badge list.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view my profile
Then I see all earned badges
```

Requirement UFR-OE-107-03: Sharing Badges on Mozilla Backpack

Overview For learners who want to showcase their badges outside the platform, this requirement integrates with Mozilla Backpack (or similar open badge services). Learners can share their earned badges to their Backpack account, making them portable and verifiable across the web.

1. **Title:** Share badges to Mozilla Backpack.
2. **Actor:** Learner.
3. **Pre:** Badge earned.
4. **Post:** Badge shared.
5. **Normal Flow:** Clicks “Share to Backpack” and authorises.
6. **Alternative Flows:** None.
7. **Business Rules:** Requires Mozilla Backpack account.
8. **Input/Output:** Input: share action. Output: badge added to Backpack.
9. **Acceptance Criteria:**

```
Given I am a learner
When I share a badge to Mozilla Backpack
Then the badge appears in my Backpack
```

0.4.8 Video Player Module

Requirement UFR-OE-108-01: Watching Videos on the Open edX Video Player

Overview The video player is the primary interface for consuming video content. This requirement defines the standard controls (play, pause, volume, speed, full-screen, closed captions) and ensures reliable playback across devices and browsers. The player must also handle adaptive bitrate streaming for smooth viewing.

1. **Title:** Use the video player.
2. **Actor:** Learner.
3. **Pre:** Video component.
4. **Post:** Video plays.
5. **Normal Flow:** Player controls: play, pause, volume, speed, full-screen, CC/subtitles.
6. **Alternative Flows:** None.
7. **Business Rules:** Video must be compatible.
8. **Input/Output:** Input: user interactions. Output: video playback.
9. **Acceptance Criteria:**

```
Given I am a learner
When I play a video
Then I can control playback and see captions if available
```

0.4.9 Discussions Module

Requirement UFR-OE-109-01: Getting Started with Discussions

Overview The discussion forum is a place for learners to ask questions, share insights, and collaborate. This requirement provides a clear entry point – the “Discussions” tab – where learners can view existing threads, filter by topic, and create new posts. The forum must be intuitive and encourage participation.

1. **Title:** Access course discussions.
2. **Actor:** Learner.
3. **Pre:** Enrolled.
4. **Post:** Discussions view.
5. **Normal Flow:** Clicks “Discussions” tab; sees threads and topics.
6. **Alternative Flows:** None.

7. Business Rules: Discussions enabled by educator.

8. Input/Output: Output: forum view.

9. Acceptance Criteria:

```
Given I am a learner
When I click "Discussions"
Then I see all threads and can create new posts
```

Requirement UFR-OE-109-02: Finding and Following Conversations

Overview To stay engaged, learners need to find relevant discussions and follow them. This requirement supports search, filtering, and a “follow” feature that sends email notifications when new replies are posted. This keeps learners informed without requiring them to constantly check the forum.

1. Title: Find and follow conversations.

2. Actor: Learner.

3. Pre: Discussions open.

4. Post: Followed thread.

5. Normal Flow: Search, filter, follow threads, get notifications.

6. Alternative Flows: None.

7. Business Rules: Following sends email notifications.

8. Input/Output: Input: follow action. Output: notification preference set.

9. Acceptance Criteria:

```
Given I am a learner
When I follow a discussion thread
Then I receive email notifications for new replies
```

Requirement UFR-OE-109-03: Taking Part in Discussions

Overview Active participation is key to a vibrant learning community. This requirement allows learners to create new threads (topics) and reply to existing ones, with all posts subject to moderation as configured by the educator. The interface must be simple and responsive.

1. Title: Participate in discussions.

2. Actor: Learner.

3. Pre: Discussions open.

4. **Post:** Post created.
5. **Normal Flow:** Creates new threads, replies to existing ones.
6. **Alternative Flows:** None.
7. **Business Rules:** Posts are moderated.
8. **Input/Output:** Input: post content. Output: visible post.
9. **Acceptance Criteria:**

```
Given I am a learner
When I reply to a thread
Then my reply appears in the discussion
```

Requirement UFR-OE-109-04: Staying Updated with Notifications

Overview To avoid missing important replies, learners receive email notifications for threads they follow or when they are mentioned. This requirement ensures that notification delivery is reliable and respects the learner's preferences (e.g., digest vs. immediate).

1. **Title:** Receive discussion notifications.
2. **Actor:** Learner.
3. **Pre:** Following threads.
4. **Post:** Notification received.
5. **Normal Flow:** System sends email notifications for followed threads.
6. **Alternative Flows:** None.
7. **Business Rules:** Notifications respect user preferences.
8. **Input/Output:** Output: email.
9. **Acceptance Criteria:**

```
Given I am a learner following a thread
When someone replies
Then I receive an email notification
```

0.4.10 Assignments & Assessments Module

Requirement UFR-OE-110-01: Completing Mathematical and Scientific Assignments

Overview Many courses involve mathematical notation. This requirement supports LaTeX input and rendering within assignments, allowing learners to write formulas and equations. The system must render the LaTeX correctly and grade the answer appropriately.

1. **Title:** Complete mathematical assignments.
2. **Actor:** Learner.
3. **Pre:** Assignment open.
4. **Post:** Answer submitted.
5. **Normal Flow:** Uses LaTeX for math expressions.
6. **Alternative Flows:** None.
7. **Business Rules:** LaTeX must be valid.
8. **Input/Output:** Input: answer with LaTeX. Output: graded answer.
9. **Acceptance Criteria:**

```
Given I am a learner
When I submit an answer with LaTeX
Then the math renders correctly and is graded
```

Requirement UFR-OE-110-02: Taking a Timed Exam

Overview Timed exams simulate real-world test conditions. This requirement provides a countdown timer that is visible throughout the exam; when time expires, the exam is automatically submitted. This ensures fairness and enforces time limits set by the educator.

1. **Title:** Take timed exam.
2. **Actor:** Learner.
3. **Pre:** Exam open.
4. **Post:** Exam submitted.
5. **Normal Flow:** Starts exam; timer counts down; auto-submit at expiration.
6. **Alternative Flows:** None.
7. **Business Rules:** Time limit set by educator.
8. **Input/Output:** Input: answers. Output: grade.

9. Acceptance Criteria:

Given I am a learner
When I start a timed exam
Then I must submit before time expires or it auto-submits

Requirement UFR-OE-110-03: Proctored Exams

Overview For high-stakes assessments, proctoring ensures integrity. This requirement integrates with a proctoring service (e.g., ProctorU) – learners must install a browser extension, verify their identity, and complete the exam under live monitoring. The system records the session for later review.

1. **Title:** Take proctored exam.
2. **Actor:** Learner.
3. **Pre:** Proctored exam assigned.
4. **Post:** Exam taken with monitoring.
5. **Normal Flow:** Installs proctoring extension; verifies identity; takes exam; monitoring active.
6. **Alternative Flows:** None.
7. **Business Rules:** Proctoring software must be installed.
8. **Input/Output:** Input: identity verification. Output: exam session.
9. **Acceptance Criteria:**

Given I am a learner
When I take a proctored exam
Then my session is monitored and recorded

Requirement UFR-OE-110-04: Explaining Multiple Choice Answers

Overview Sometimes a learner wants to justify their choice. This optional feature allows learners to provide a brief explanation alongside their answer. This can be useful for instructors to understand reasoning and for learners to reflect on their decisions.

1. **Title:** Provide explanation for multiple choice answers.
2. **Actor:** Learner.
3. **Pre:** Multiple choice question.
4. **Post:** Explanation saved.
5. **Normal Flow:** Selects answer and optionally enters explanation.

6. **Alternative Flows:** None.
7. **Business Rules:** Explanation optional.
8. **Input/Output:** Input: explanation. Output: stored with answer.
9. **Acceptance Criteria:**

```
Given I am a learner
When I select an answer and add an explanation
Then the explanation is stored for review
```

Requirement UFR-OE-110-05: Completing Essay Assignments

Overview Essay assignments allow open-ended responses. This requirement provides a rich text editor where learners can write, format, and submit their essays. The system stores the submission and makes it available for grading by instructors or peers.

1. **Title:** Complete essay assignment.
2. **Actor:** Learner.
3. **Pre:** Assignment open.
4. **Post:** Essay submitted.
5. **Normal Flow:** Writes essay in text editor (supports formatting).
6. **Alternative Flows:** None.
7. **Business Rules:** Minimum length may be set.
8. **Input/Output:** Input: essay text. Output: submission confirmation.
9. **Acceptance Criteria:**

```
Given I am a learner
When I submit an essay
Then it is saved and available for grading
```

Requirement UFR-OE-110-06: Open Response Assessment Steps

Overview The ORA workflow is multi-step: submit, self-assess, peer-assess, and receive score. This requirement guides learners through these steps with clear instructions and progress indicators, ensuring they complete each phase correctly.

1. **Title:** Follow ORA workflow.
2. **Actor:** Learner.
3. **Pre:** ORA assigned.

4. **Post:** Assessment completed.
5. **Normal Flow:** Steps: submit response, self-assessment, peer assessment, receive score.
6. **Alternative Flows:** None.
7. **Business Rules:** Order of steps fixed.
8. **Input/Output:** Input: response, assessments. Output: final score.
9. **Acceptance Criteria:**

```
Given I am a learner
When I complete all ORA steps
Then I receive a final score
```

Requirement UFR-OE-110-07: How Grading Is Done In Open Response Assessments

Overview To demystify the grading process, this requirement provides transparency about the rubric and how peer scores are combined. Learners can view the grading criteria and understand how their final score is calculated, building trust in the system.

1. **Title:** Understand ORA grading.
2. **Actor:** Learner.
3. **Pre:** ORA available.
4. **Post:** Grading explanation shown.
5. **Normal Flow:** System displays grading rubric and peer review scores.
6. **Alternative Flows:** None.
7. **Business Rules:** Rubric defined by educator.
8. **Input/Output:** Output: grading description.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view the ORA grading criteria
Then I see the rubric and how scores are calculated
```

Requirement UFR-OE-110-08: Completing an Open Response Assessment

Overview This requirement covers the full ORA completion cycle. Learners submit their response, perform self-assessment, review peers, and finally see their aggregated score. Each step must be intuitive, with clear deadlines and reminders.

1. **Title:** Complete ORA.
2. **Actor:** Learner.
3. **Pre:** ORA open.
4. **Post:** ORA completed.
5. **Normal Flow:** Submits response, performs self/peer assessments.
6. **Alternative Flows:** None.
7. **Business Rules:** Peer assessment requires reviewing others.
8. **Input/Output:** Input: response, peer reviews. Output: final score.
9. **Acceptance Criteria:**

```
Given I am a learner
When I complete an ORA
Then I see my final grade and feedback
```

Requirement UFR-OE-110-09: Receive Score and Provide Feedback

Overview After an ORA is graded, learners can view their score and, if desired, provide feedback on the assessment experience. This feedback loop helps improve future assessments and gives learners a voice in the process.

1. **Title:** Receive ORA score and provide feedback.
2. **Actor:** Learner.
3. **Pre:** Assessment done.
4. **Post:** Score displayed.
5. **Normal Flow:** Views score and provides feedback on assessment experience.
6. **Alternative Flows:** None.
7. **Business Rules:** Feedback optional.
8. **Input/Output:** Input: feedback. Output: stored.
9. **Acceptance Criteria:**

```
Given I am a learner
When I receive my ORA score
Then I can optionally submit feedback on the process
```

Requirement UFR-OE-110-10: How Peer Assessment Scores Are Calculated

Overview Peer assessment scoring can be complex. This requirement explains the algorithm (e.g., average, median, or adjusted scores) so learners understand how multiple peer evaluations are combined into a final score. This transparency reduces confusion and increases acceptance of peer grading.

1. **Title:** Understand peer assessment scoring.
2. **Actor:** Learner.
3. **Pre:** ORA with peer assessment.
4. **Post:** Scoring explained.
5. **Normal Flow:** System explains scoring algorithm (average, median, adjusted).
6. **Alternative Flows:** None.
7. **Business Rules:** Outliers may be removed.
8. **Input/Output:** Output: scoring description.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view how peer scores are calculated
Then I understand the algorithm used
```

Requirement UFR-OE-110-11: Canceled Responses

Overview Sometimes a learner may submit a response and later wish to withdraw it before any assessment has started. This requirement allows cancellation, giving learners the flexibility to correct errors or change their mind. Once assessment begins, cancellation is not permitted.

1. **Title:** Handle canceled responses.
2. **Actor:** Learner.
3. **Pre:** Response submitted.
4. **Post:** Cancellation confirmed.
5. **Normal Flow:** Can cancel submission before assessment.
6. **Alternative Flows:** None.
7. **Business Rules:** Cancellation allowed before assessment phase.
8. **Input/Output:** Input: cancel action. Output: submission removed.
9. **Acceptance Criteria:**

```
Given I am a learner
When I cancel my submission before assessment
Then I can resubmit later
```

0.4.11 Team Projects Module

Requirement UFR-OE-111-01: About Teams and Topics

Overview Team projects encourage collaboration. This requirement provides an introductory page that explains what teams are, how topics (project themes) work, and how learners can form or join teams. This sets the stage for productive group work.

1. **Title:** Understand teams and topics.
2. **Actor:** Learner.
3. **Pre:** Course with team feature.
4. **Post:** Teams explained.
5. **Normal Flow:** System provides team and topic overview.
6. **Alternative Flows:** None.
7. **Business Rules:** Topics defined by educator.
8. **Input/Output:** Output: description.
9. **Acceptance Criteria:**

```
Given I am a learner
When I view the Teams page
Then I see an explanation of teams and topics
```

Requirement UFR-OE-111-02: Search for a Team

Overview To find a suitable team, learners need search functionality. This requirement allows searching by team name, topic, or member name, making it easy to discover existing teams. Search results are case-insensitive and ordered by relevance.

1. **Title:** Search for a team.
2. **Actor:** Learner.
3. **Pre:** Teams page.
4. **Post:** Search results.
5. **Normal Flow:** Searches teams by name, topic, or member.
6. **Alternative Flows:** None.
7. **Business Rules:** Search is case-insensitive.
8. **Input/Output:** Input: search term. Output: list of teams.
9. **Acceptance Criteria:**

```
Given I am a learner
When I search for a team name
Then I see matching teams
```

Requirement UFR-OE-111-03: Join a Team

Overview Joining a team may require approval, depending on the team's settings. This requirement provides a "Join Team" button; if approval is required, the team captain receives a request. Once accepted, the learner becomes a full member with access to team resources.

1. **Title:** Join an existing team.
2. **Actor:** Learner.
3. **Pre:** Team visible.
4. **Post:** Added to team.
5. **Normal Flow:** Clicks "Join Team"; may require approval.
6. **Alternative Flows:** None.
7. **Business Rules:** Team size may be limited.
8. **Input/Output:** Input: join request. Output: membership.
9. **Acceptance Criteria:**

```
Given I am a learner
When I join a team
Then I become a member and can access team discussions
```

Requirement UFR-OE-111-04: Leave a Team

Overview Learners should be able to leave a team if they are no longer interested or if they need to switch. This requirement provides a "Leave Team" button; if the leaving member is the last member, the team may be dissolved automatically to avoid orphaned groups.

1. **Title:** Leave a team.
2. **Actor:** Learner.
3. **Pre:** Member of team.
4. **Post:** Removed from team.
5. **Normal Flow:** Clicks "Leave Team" and confirms.
6. **Alternative Flows:** None.
7. **Business Rules:** Cannot leave if only member (may dissolve team).
8. **Input/Output:** Input: leave action. Output: removed.
9. **Acceptance Criteria:**

```
Given I am a team member
When I leave the team
Then I am removed and lose access to team resources
```

Requirement UFR-OE-111-05: Create a Team

Overview If no suitable team exists, learners can create their own. This requirement provides a form to enter team name, description, topic, and enrollment type (open or by approval). The team name must be unique within the course to avoid duplication.

1. **Title:** Create a new team.
2. **Actor:** Learner.
3. **Pre:** Course allows team creation.
4. **Post:** Team created.
5. **Normal Flow:** Enters name, description, topic, enrollment type; clicks Create.
6. **Alternative Flows:** None.
7. **Business Rules:** Team name unique within course.
8. **Input/Output:** Input: team details. Output: new team.
9. **Acceptance Criteria:**

```
Given I am a learner
When I create a team with a unique name
Then the team appears in the team list
```

Requirement UFR-OE-111-06: Participating in Team Discussions

Overview Team members need a private space to communicate. This requirement provides a team-specific discussion forum where only members can view and post. This facilitates coordination, task assignment, and sharing of resources within the team.

1. **Title:** Participate in team discussions.
2. **Actor:** Learner.
3. **Pre:** Member of team.
4. **Post:** Discussion posted.
5. **Normal Flow:** Team members discuss within team-specific forum.
6. **Alternative Flows:** None.
7. **Business Rules:** Team discussions are private.
8. **Input/Output:** Input: post. Output: visible to team.
9. **Acceptance Criteria:**

```
Given I am a team member
When I post in team discussion
Then only team members can see my post
```

0.4.12 Search Module

Requirement UFR-OE-112-01: Overview of Search

Overview To help learners find content efficiently, the platform provides a full-text search powered by Meilisearch. This requirement ensures that search help is available, explaining what is searchable (course titles, content, discussions) and how to refine queries.

1. **Title:** Understand search capabilities.
2. **Actor:** Learner.
3. **Pre:** None.
4. **Post:** Search explained.
5. **Normal Flow:** System provides help text about full-text search.
6. **Alternative Flows:** None.
7. **Business Rules:** Search powered by Meilisearch.
8. **Input/Output:** Output: help text.
9. **Acceptance Criteria:**

```
Given I am a learner
When I look for search help
Then I see a description of what is searchable
```

Requirement UFR-OE-112-02: Search a Single Course

Overview When inside a course, learners often want to find content within that course only. This requirement provides a search box that restricts results to the current course, making it easy to locate specific topics without being overwhelmed by other courses.

1. **Title:** Search within a course.
2. **Actor:** Learner.
3. **Pre:** In a course.
4. **Post:** Results displayed.
5. **Normal Flow:** Enters query; returns relevant content within course.
6. **Alternative Flows:** None.
7. **Business Rules:** Search scope is the current course.
8. **Input/Output:** Input: query. Output: results.
9. **Acceptance Criteria:**

```
Given I am a learner in a course
When I search for a term
Then I see results only from that course
```

Requirement UFR-OE-112-03: Search All Your Courses

Overview From the dashboard, learners may want to search across all their enrolled courses. This requirement provides a global search that returns results from all courses the learner is enrolled in, helping them quickly find content without navigating each course individually.

1. **Title:** Search across all courses.
2. **Actor:** Learner.
3. **Pre:** On dashboard.
4. **Post:** Results displayed.
5. **Normal Flow:** Enters query; returns results from all enrolled courses.
6. **Alternative Flows:** None.
7. **Business Rules:** Search scope is all enrolled courses.
8. **Input/Output:** Input: query. Output: results.
9. **Acceptance Criteria:**

Given I am a learner on the dashboard
When I search for a term
Then I see results from all my courses

0.4.13 Bookmarking Module

Requirement UFR-OE-113-01: Add or Remove a Bookmark

Overview Bookmarks allow learners to save important content for quick reference. This requirement provides a bookmark icon on each piece of content; clicking it toggles the bookmark status. Bookmarked content is then listed in a dedicated bookmarks page for easy access.

1. **Title:** Bookmark course content.
2. **Actor:** Learner.
3. **Pre:** Viewing content.
4. **Post:** Bookmark toggled.
5. **Normal Flow:** Clicks bookmark icon to save/remove bookmark.
6. **Alternative Flows:** None.
7. **Business Rules:** One bookmark per user per content.
8. **Input/Output:** Input: click icon. Output: bookmark status.

9. Acceptance Criteria:

Given I am a learner viewing content
When I click the bookmark icon
Then the content is bookmarked and appears in my bookmarks list

Requirement UFR-OE-113-02: View Your Bookmarks

Overview The bookmarks page provides a consolidated list of all saved content. Bookmarks are sorted by date added, and learners can click on any entry to navigate directly to that content. This serves as a personal study guide.

1. **Title:** View bookmarked content.
2. **Actor:** Learner.
3. **Pre:** Bookmarks exist.
4. **Post:** List displayed.
5. **Normal Flow:** Accesses bookmarks page to view and navigate saved items.
6. **Alternative Flows:** None.
7. **Business Rules:** Sorted by date added.
8. **Input/Output:** Output: bookmark list.
9. **Acceptance Criteria:**

Given I have bookmarks
When I view my bookmarks
Then I see a list of all bookmarked content

0.4.14 Notifications & Preferences Module

Requirement UFR-OE-114-01: Where Notifications Appear

Overview Notifications keep learners informed about important events (replies, due dates, updates). This requirement defines that notifications appear in three places: the dashboard, email, and in-app alerts (bell icon). The bell icon shows a badge count for unread notifications.

1. **Title:** Access notifications.
2. **Actor:** Learner.
3. **Pre:** Logged in.
4. **Post:** Notifications visible.
5. **Normal Flow:** Notifications appear in dashboard, email, and in-app alerts.

- 6. **Alternative Flows:** None.
- 7. **Business Rules:** In-app via bell icon.
- 8. **Input/Output:** Output: notification display.
- 9. **Acceptance Criteria:**

```
Given I am a learner
When I have unread notifications
Then I see the bell icon with a badge count
```

Requirement UFR-OE-114-02: Activity Types

Overview To help learners understand what triggers notifications, this requirement lists all activity types: discussion replies, assignment due dates, course updates, and feedback on assessments. This transparency helps learners manage their expectations and preferences.

- 1. **Title:** Understand notification activity types.
- 2. **Actor:** Learner.
- 3. **Pre:** None.
- 4. **Post:** Types explained.
- 5. **Normal Flow:** System lists notification types (discussion replies, due dates, updates, feedback).
- 6. **Alternative Flows:** None.
- 7. **Business Rules:** Types defined by system.
- 8. **Input/Output:** Output: list of types.
- 9. **Acceptance Criteria:**

```
Given I am a learner
When I view notification settings
Then I see all available activity types
```

Requirement UFR-OE-114-03: Managing Preferences

Overview Learners have different preferences for how and when they receive notifications. This requirement provides a settings panel where learners can enable/disable notifications per activity type and choose delivery channels (email, in-app). Preferences are stored per user.

- 1. **Title:** Manage notification preferences.
- 2. **Actor:** Learner.

3. **Pre:** Logged in.
4. **Post:** Preferences saved.
5. **Normal Flow:** Configures which notifications to receive and via which channels.
6. **Alternative Flows:** None.
7. **Business Rules:** Preferences stored per user.
8. **Input/Output:** Input: settings. Output: saved.
9. **Acceptance Criteria:**

```
Given I am a learner
When I change notification preferences
Then my preferences are updated
```

Requirement UFR-OE-114-04: Notifications Expiry

Overview To keep the notification list manageable, this requirement defines that notifications older than 30 days are automatically archived. Archived notifications are still accessible but not in the active inbox, reducing clutter.

1. **Title:** Notification expiry.
2. **Actor:** System.
3. **Pre:** Notification older than 30 days.
4. **Post:** Notification archived.
5. **Normal Flow:** Notifications expire after 30 days.
6. **Alternative Flows:** None.
7. **Business Rules:** Archived notifications can still be viewed.
8. **Input/Output:** None.
9. **Acceptance Criteria:**

```
Given a notification is 30 days old
When the system runs the expiry job
Then the notification is moved to archive
```

0.4.15 Course Wiki Module

Requirement UFR-OE-115-01: Using the Course Wiki

Overview The course wiki is a collaborative space where learners can contribute knowledge, share resources, and build a living document. This requirement provides access to the wiki from the course navigation, with edit permissions controlled by the educator. Learners can view and, if allowed, edit pages.

1. **Title:** Use the course wiki.
2. **Actor:** Learner.
3. **Pre:** Course has wiki enabled.
4. **Post:** Wiki displayed.
5. **Normal Flow:** Accesses wiki from course navigation; views and edits (if allowed).
6. **Alternative Flows:** None.
7. **Business Rules:** Edit permissions controlled by educator.
8. **Input/Output:** Input: edit (if allowed). Output: wiki page.
9. **Acceptance Criteria:**

Given I am a learner
When I open the course wiki
Then I can view and optionally edit pages

Requirement UFR-OE-115-02: View Live Course

Overview This requirement ensures that learners only see published and released content – content that the educator has made visible. Unpublished content is hidden, and learners are not confused by draft materials. This maintains a clear and reliable learning path.

1. **Title:** View live course content.
2. **Actor:** Learner.
3. **Pre:** Content published.
4. **Post:** Content visible.
5. **Normal Flow:** Accesses content that is currently active/published.
6. **Alternative Flows:** None.
7. **Business Rules:** Only published and released content is visible.
8. **Input/Output:** Input: navigation. Output: content.

9. Acceptance Criteria:

Given I am a learner
When I view a course
Then I only see content that has been published and released

0.5 Security and Compliance Requirements (Non-Functional)

This section defines mandatory security requirements derived from INSA's CMCSRS, the Computer Crime Proclamation, and the Electronic Signature Proclamation. All requirements are traceable to the *Security Architecture & Threat Model* document.

0.5.1 Access Control (CMCSRS 6.17)

- SEC-NFR-001-1.** The system must enforce role-based access control (RBAC) for all user actions.
- SEC-NFR-001-2.** All access to administrative functions must be restricted to users with appropriate privileges.
- SEC-NFR-001-3.** User accounts must be subject to a formal provisioning and de-provisioning process.
- SEC-NFR-001-4.** Default passwords for any system component must be changed upon first use.
- SEC-NFR-001-5.** The system must implement multi-factor authentication (MFA) for administrative accounts.

0.5.2 Cryptography (CMCSRS 6.18)

- SEC-NFR-002-1.** All data in transit must be encrypted using TLS 1.2 or higher.
- SEC-NFR-002-2.** All sensitive data at rest must be encrypted using strong, INSA-approved algorithms.
- SEC-NFR-002-3.** Private keys and secrets must never be stored in plaintext.
- SEC-NFR-002-4.** Digital signatures must be used for certificate verification as per the Electronic Signature Proclamation.

0.5.3 Network Security (CMCSRS 5.7.c)

- SEC-NFR-003-1.** A host-based firewall must be active and configured to allow only necessary ports.
- SEC-NFR-003-2.** The internal network must be segmented into separate Docker networks.

SEC-NFR-003-3. All unnecessary services and ports must be disabled.

SEC-NFR-003-4. Intrusion detection/prevention systems must be deployed.

0.5.4 Logging and Monitoring (CMCSRS 6.25)

SEC-NFR-004-1. All logs must be collected centrally and retained for at least one month.

SEC-NFR-004-2. Security-relevant events must be logged with timestamps and user identifiers.

SEC-NFR-004-3. Regular vulnerability scanning must be performed weekly.

SEC-NFR-004-4. Anomaly detection and alerting must be configured.

0.5.5 Incident Response and Business Continuity (CMCSRS 6.26, 6.19)

SEC-NFR-005-1. An incident response plan must be documented and tested annually.

SEC-NFR-005-2. A business continuity plan with defined RTO and RPO must be maintained.

SEC-NFR-005-3. Regular backups must be performed and stored off-site.

SEC-NFR-005-4. All security incidents must be reported to Ethio-CERT and INSA.

0.5.6 Personnel Security (CMCSRS 5.6, 6.37)

SEC-NFR-006-1. All personnel with access to sensitive systems must undergo security clearance.

SEC-NFR-006-2. Mandatory annual security awareness training must be provided.

SEC-NFR-006-3. Job descriptions must include security responsibilities.

0.5.7 Compliance and Audit (CMCSRS 6.21, 6.22, 6.23)

SEC-NFR-007-1. The system must undergo an annual security audit by an INSA-certified auditor.

SEC-NFR-007-2. All information systems must be accredited before deployment.

SEC-NFR-007-3. A compliance report must be submitted to INSA annually.

0.6 Non-Functional Requirements

0.6.1 Performance

- Support at least 10,000 concurrent learners and 500 concurrent educators.
- Page load times below 2 seconds for 95% of requests.
- API response times below 500ms for 95% of requests.
- Video streaming supports at least 720p with adaptive bitrate.

0.6.2 Security

- All network communication encrypted with TLS 1.2+.
- Passwords hashed using PBKDF2/BCrypt.
- API authentication using JWT with short expiry.
- RBAC enforced.
- Firewall and intrusion detection in place.

0.6.3 Availability

- 99.9% uptime.
- Automatic container restarts.
- Daily off-site backups.

0.6.4 Maintainability

- Containerised microservices architecture.
- Quarterly upstream merges.
- Externalised configuration.

0.6.5 Scalability

- Horizontal scaling with container replicas.
- Database read replicas supported.

0.6.6 Usability

- Responsive interface (modern browsers).
- User-friendly error messages.
- WCAG 2.1 AA accessibility.

0.7 Traceability Matrix

Requirement ID	Test Case ID	Security Reference
UFR-OE-001-01	TC-001	SEC-NFR-001, SEC-NFR-002
UFR-OE-001-03	TC-002	SEC-NFR-001
UFR-OE-003-01	TC-003	SEC-NFR-003, SEC-NFR-004
UFR-OE-003-02	TC-004	SEC-NFR-004
UFR-OE-004-01	TC-005	SEC-NFR-002
UFR-OE-004-03	TC-006	SEC-NFR-002
UFR-OE-005-03	TC-007	SEC-NFR-001, SEC-NFR-002
UFR-OE-006-03	TC-008	SEC-NFR-001, SEC-NFR-004
UFR-OE-103-01	TC-009	SEC-NFR-001
UFR-OE-106-03	TC-010	—
UFR-OE-107-01	TC-011	SEC-NFR-002
UFR-OE-109-01	TC-012	SEC-NFR-001, SEC-NFR-003
UFR-OE-110-02	TC-013	SEC-NFR-003
UFR-OE-112-02	TC-014	—
UFR-OE-113-01	TC-015	—
UFR-OE-114-03	TC-016	SEC-NFR-006
SEC-NFR-001..007	Security doc	CMCSRS, Proc.958/2016

0.8 Compliance Matrix

The following table maps system requirements to specific INSA regulations.

INSA Regulation	Section	System Compliance Evidence
CMCSRS v2.0	5.3 Leadership	Top leadership allocated 0.5% budget; strategic awareness program
CMCSRS v2.0	5.4 Governance	Security committee established; CISO appointed
Continued		

INSA Regulation	Section	System Compliance Evidence
CMCSRS v2.0	5.5 Management	Risk-based approach implemented; comprehensive risk profiles
CMCSRS v2.0	5.7 Technology	Firewall, network segmentation, vulnerability scanning
CMCSRS v2.0	6.4 Risk Assessment	SGOC+PESTLE, BMIS, ISO 27001 assessments
CMCSRS v2.0	6.17 Access Control	RBAC enforced; non-root user; secrets rotated
CMCSRS v2.0	6.18 Cryptography	TLS 1.2; AES-256; JWT RSA keys rotated
CMCSRS v2.0	6.25 Monitoring	Trivy daily scans; centralised logging; health checks
CMCSRS v2.0	6.26 Incident Response	Fail2ban; IRP documented; reporting to Ethio-CERT
CMCSRS v2.0	6.28 Backup	Daily automated backups; retention policy
Computer Crime Proc.	Art 3-8	Access controls; firewall; logging; incident detection
Electronic Signature Proc.	Art 5-8	Digital signatures; certificate verification
National Cyber Policy	2.2 Legal Framework	Compliance with all applicable laws
National Cyber Policy	2.3 Awareness	Security awareness program established
National Cyber Policy	2.4 Capacity Building	Staff training; career path alignment

0.9 Glossary

CMCSRS Critical Mass Cyber Security Requirement Standard.

CMS Course Management System.

DFD Data Flow Diagram.

INSA Information Network Security Administration.

LMS Learning Management System.

MFE Micro-Frontend.

RBAC Role-Based Access Control.

Tutor Open edX deployment tool.

0.10 References

- *Security Architecture & Threat Model – Moove Education Platform* (Security_OpenEdX.tex)
- CMCSRS v2.0 (2025)
- FDRE National Cybersecurity Policy (2024)
- Computer Crime Proclamation No. 958/2016
- Electronic Signature Proclamation No. 1072/2018
- Cyber Security Audit and Evaluation Guideline v1.0
- Cyber Security Risk Assessment Framework v1.0
- Swagger API Documentation (swagger-docs.json)
- Deployment Report (deployment-report.txt)
- VPS Health Check (vps_health_check.txt)

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